Towards Non-Invasive Breslow Staging: A Multitask Vision Transformer with Self-Supervised Pretraining for Melanoma Prognosis

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***Abstract*—Accurate assessment of melanoma malignancy and Breslow thickness is essential for prognosis and treatment planning, yet thickness measurement requires invasive biopsy and histopathological examination. This work presents an image-only deep learning framework for simultaneous melanoma classification and continuous Breslow thickness estimation from dermoscopic images. A self-supervised DINOv2 Vision Transformer backbone is first domain-adapted using a large corpus of unlabeled dermoscopic images following the DermINO pretraining paradigm, and subsequently integrated into a dual-head multitask architecture. The model is trained on a curated dataset of 3,119 dermoscopic images (1,104 melanoma and 2,015 benign), including 499 cases with thickness annotations. To address severe class and label imbalance, we employ focal loss, class-aware augmentation, stratified splitting, and weighted multitask optimization. On held-out data, the proposed framework achieves an AUROC of 0.967 for melanoma classification and a mean absolute error of 0.137 mm (R^2 = 0.847) for thickness prediction, demonstrating reliable malignancy discrimination and clinically meaningful sub-millimeter prognostic accuracy. The results indicate that domain-adapted self-supervised transformers can enable accurate, interpretable, and non-invasive melanoma prognosis using dermoscopic imagery alone.**

I. Introduction

Cutaneous melanoma is among the most aggressive forms of skin cancer, with patient survival strongly dependent on early detection and accurate staging. The most critical prognostic factor in melanoma assessment is Breslow thickness, which measures tumor invasion depth and directly determines the American Joint Committee on Cancer (AJCC) staging category [18], surgical margin selection, and eligibility for sentinel lymph node biopsy. However, Breslow thickness can only be measured following invasive excision and histopathological analysis, limiting its utility for early risk stratification and non-invasive clinical decision support.

Recent advances in deep learning have significantly improved automated melanoma classification from dermoscopic images. Vision Transformers (ViTs), in particular, have demonstrated strong performance by capturing long-range spatial dependencies across lesion structures [8], [9], while GAN-based synthetic augmentation has been employed to mitigate the chronic class imbalance inherent to melanoma datasets [8], [12]. Nevertheless, most existing approaches focus exclusively on binary malignancy detection. Relatively few studies address continuous prognostic estimation such as Breslow thickness prediction, and those that do tend to formulate the problem as coarse categorical classification rather than true regression [7]. Santaniello et al. [1] identified several critical limitations in this space: small dataset sizes, disproportionate class distributions, limited model interpretability, and the absence of multimodal prognostic cues.

Self-supervised learning has emerged as a promising paradigm to address limited annotation availability by leveraging large unlabeled datasets to learn transferable feature representations [14]. Xu et al. [2] demonstrated that domain-specific DINOv2 pretraining on dermoscopic images yields accuracy gains of 5–8% over ImageNet-pretrained models for melanoma classification, confirming the advantage of aligning feature representations to the statistical properties of skin lesion imagery prior to supervised fine-tuning. Multimodal fusion frameworks [3]–[6] have further shown that cross-attention architectures can integrate image and clinical metadata to improve diagnostic performance, though these systems concentrate on classification and do not address continuous thickness regression.

Despite these advances, several gaps remain in the literature. No existing study integrates domain-adapted self-supervised transformer pretraining with a dual-head multitask architecture capable of simultaneous melanoma classification and continuous Breslow thickness regression [1], [7]. Moreover, robust evaluation standards—including stratified data splits, reproducible preprocessing pipelines, and principled imbalance handling—are inconsistently applied across published work.

To address these limitations, this work proposes an image-only multitask framework that jointly performs melanoma classification and continuous Breslow thickness estimation using a domain-adapted DINOv2 Vision Transformer backbone. The model is trained on a curated dataset combining multiple dermoscopic sources, supported by a reproducible data processing pipeline and comprehensive imbalance handling strategies combining focal loss [20], class-aware augmentation, and

weighted regression loss.

The main contributions of this work are as follows:

- An image-only multitask framework that simultaneously performs melanoma classification and continuous Breslow thickness regression.
- Domain-specific self-supervised pretraining using a DINOv2 Vision Transformer [14] adapted on large-scale unlabeled dermoscopic data following the DermINO methodology [2].
- A robust data preparation and training pipeline incorporating stratified splitting, class-aware augmentation, and focal-loss-based [20] imbalance handling.
- Demonstration of clinically meaningful performance, achieving high classification accuracy and sub-millimeter thickness estimation precision within AJCC staging boundaries [18].

II. Related Work

Artificial Neural Networks, Deep learning progress in Vision Transformer, self-supervised learning, and multimodal fusion architectures has seen rapid development in automated Melanoma diagnosis and prognosis. This section summarizes the recent research in four major categories, including self-supervised models, multimodal fusion strategies, Vision Transformer to analyse skin lesions, and Breslow thickness prediction.

A. Foundational Self-Supervised Models and Pretraining

The limited availability of labeled medical images has spawned self-supervised learning algorithms that use large unlabeled datasets. DINO (self-DIstillation with NO labels) [14] is an unlabeled knowledge distillation of student and teacher networks to learn rich visual representations. Xu et al. [2] presented DermINO, a hybrid pretraining model based on the fusion of DINOv2 self-supervised training and supervised fine-tuning applied to dermoscopic images. DermINO gives higher accuracy gains of 5–8 percent over ImageNet-pretrained models on melanoma classification problems on ISIC, HAM10000, and in-house data. The work has demonstrated that domain-specific pretraining outperforms general-purpose ImageNet representations at fine-grained dermoscopic features (vascular patterns, pigment networks).

We take this pattern a step further and apply the DINO pretraining on 14,000+ unlabeled dermoscopic images and later fine-tune our tailored multi-source dataset to predict melanoma prognosis.

B. Melanoma Classification with Vision Transformers

ViTs have become competitive with CNNs on skin lesion analysis since they are capable of capturing long-range dependencies by means of self-attention. ViTs divide images into patches that do not overlap and process them with a series of transformer encoder layers to capture global information.

Garcia et al. [8] have performed a multi-dataset analysis of ViT-L/16 to classify melanoma with an average AUC of 0.915 on ISIC, HAM10000, and PH2. Their contribution focuses on the concept of aggressive data augmentation, whereby the authors use GAN-based synthesis to equalize class ratios by synthesizing melanoma samples. Likewise, Wang et al. [12] used ViT together with synthetic images made by StyleGAN2, which raised the representation of minority classes by 300 percent and obtained 98.43 percent accuracy on ISIC 2020. Hossain [9] investigated custom ViT architectures using focal loss to balance the classes and achieved 93.43% on BCN20000. It was discovered that smaller patch sizes (8×8 vs. 16×16) are more effective in recognition of fine-grained texture in melanoma-specific features.

Although these studies prove ViT effectiveness in melanoma classification, they concentrate on unimodal image-only learning with ImageNet pretraining. We overcome this shortcoming by using domain-specific DINO pretraining.

C. Multimodal Image-Clinical Metadata Fusion

Dermoscopic images integrated with structured clinical characteristics (age, sex, lesion location, Breslow thickness, AJCC stage) have attracted attention for their complementary prognostic information relative to visual appearance alone. Recent works examine different fusion architectures that have attention mechanisms as the most favorable.

Tran-Van and Le [3] introduced a cross-attention fusion system for multimodal skin lesion classification with a 95.73% accuracy on HAM10000. Their two-way attention system allows image features to converse with clinical features and vice versa, explicitly learning inter-modality correlations. A systematic comparison of fusion strategies (cross-attention, self-attention, concatenation, gated fusion) on HAM10000 by Das et al. [5] showed that cross-attention is consistently 3–5% more accurate than simple concatenation, especially on minority classes.

Panneerselvam et al. [4] used attention-based fusion with transfer learning, obtaining an accuracy of 98.69% on ISIC, and solved the class imbalance problem by weighted loss functions. Li et al. [6] proposed HyperFusionNet, a cross-attention multi-path ViT that learns two tasks (classification and segmentation) with an accuracy of 93.24% on ISIC.

These studies find that cross-attention fusion is better than concatenation-based methods. Nevertheless, they are more concerned with classification and do not address continuous Breslow thickness prediction or domain-specific self-supervised pretraining. Our work addresses these gaps through a unified image-only multitask framework combining DINO pretraining with joint classification and thickness regression.

D. Breslow Thickness Prediction and Prognosis

Breslow thickness (BT) is the most important prognostic factor in melanoma staging and treatment planning. Nonetheless, the automated prediction of BT from dermoscopic images remains difficult because of the lack of labelled data and weak visual cues for depth information.

Santaniello et al. [1] carried out a systematic review of AI models for estimating BT and observed several critical challenges: (1) current datasets are small (most have fewer

than 2,000 BT-marked samples), (2) class distributions are disproportionate, (3) lack of interpretability limits clinical trust, and (4) lack of multimodality in combining AJCC stage, mitotic rate, and ulceration state. Lallas et al. [7] conducted a narrative review of machine learning algorithms for melanoma prognosis, noting that most studies describe binary thickness classification (e.g., $\leq$1 mm vs. $>$1 mm) rather than continuous regression. The authors support semi-supervised and self-supervised learning to leverage large unlabeled dermoscopic image collections for improved prognostic modelling.

By (1) predicting BT as continuous regression with MAE as an evaluation metric, (2) pretraining without labels as recommended by [7], and (3) providing attention map visualisations for interpretability as emphasised by [1], we fill in these gaps.

E. Class Imbalance on Melanoma Data

Melanoma datasets are highly class imbalanced, with benign lesions accounting for 85–90 percent of the samples. This disproportion worsens for advanced stages and thick melanomas, which are rare but clinically significant. Recent work uses multi-pronged approaches consisting of loss reweighting, data augmentation, and synthetic generation.

Focal loss [20] down-weights well-classified examples with a modulating factor $(1-p_t)^\gamma$. Panneerselvam et al. [4], Wang et al. [12], and Hossain [9] all credit 10–15% improvements in minority class recall with focal loss over standard cross-entropy. GAN-based synthesis was used in several works [8], [12] to balance class distributions. Das et al. [5] found that combining class-balanced sampling with focal loss achieves optimal performance on minority classes.

We employ a hybrid combination of weighted multi-task loss, class-balanced augmentation with stronger minority transforms, stratified train/val/test splits, and focal loss ($\gamma = 2$), incorporating the best practices found in [4], [5], [8], [9], [12].

F. Research Gaps and Contributions

There are significant gaps in the existing literature: (1) no previous study has integrated domain-specific DINO pretraining with a dual-head multitask architecture for simultaneous melanoma classification and continuous BT regression, (2) multimodal procedures [3]–[6] target classification with no continuous BT regression, (3) BT prediction experiments [1], [7] are coarse categorization-based, and (4) evaluation procedures are isolated with disparate preprocessing and data partitions.

These gaps are systematically filled by our system that has: (1) domain-specific DINO pretraining on 14,000+ unlabeled dermoscopic images using the methodology of DermINO [2], (2) joint classification and continuous BT regression following multi-task paradigms [6], and (3) extensive class imbalance control combining strategies from [4], [5], [8], [9], [12]. This is, to the best of our knowledge, the first attempt to consolidate these elements into a unified image-only melanoma prognosis framework.

III. Methods

A. Dataset Overview

This study utilizes Dermoscopic Images from different publicly available imaging datasets including ISIC Archive and Dermoscopic Atlas. The combined dataset comprised of 3,119 records which includes 2,015 benign lesions and 1,104 melanoma cases, resulting in a moderate class imbalance with a benign-to-melanoma ratio of approximately 1.8:1. The Fig. 1 illustrates the sources and their contributions; most of these cases are early stage (AJCC stage 0), which is consistent with the trend in these public skin image banks to focus on early resolution. Each image was normalized and set to 224$\times$224 before they were used for training. We split the data into training, validation and testing groups along with special attention given to class distribution to maintain the same balance of classes and the same mix of image sources. For the prognosis task we made use of 1,104 melanoma images with the thickness label, and for the benign images the thickness was simply set to zero so the model could use this as a baseline to train the thickness estimation branch.

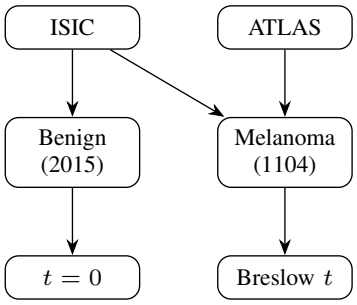


Fig. 1. Overview of data sourcing and thickness assignment.

B. Domain Adaptation via Self-Supervised Learning

To help the model handle different image sources and make better use of the limited labeled material, we first turned to self-supervised training, where we applied Vision Transformer (DINOv2) to a separate stack of 14,738 unlabeled Dermoscopic images from the ISIC archive that were not part of the labeled set. At this stage, the network practiced spotting what remained unchanged in photos, whether they were flipped, cropped, or retouched—without the need for handwritten labels at all. This training allowed the model to sense patterns in skin texture, uneven tone and general appearance of lesions, so that by the time it was asked to differentiate between benign and malignant and estimate thickness, it had a "look and feel" of the skin images. After learning from a large batch of unlabeled images, the core of this network was connected to a multitasking setup used to classify melanoma and measure Breslow thickness.

C. Model Architecture

This method utilizes Vision Transformer model as the backbone upon which two fully connected network heads are

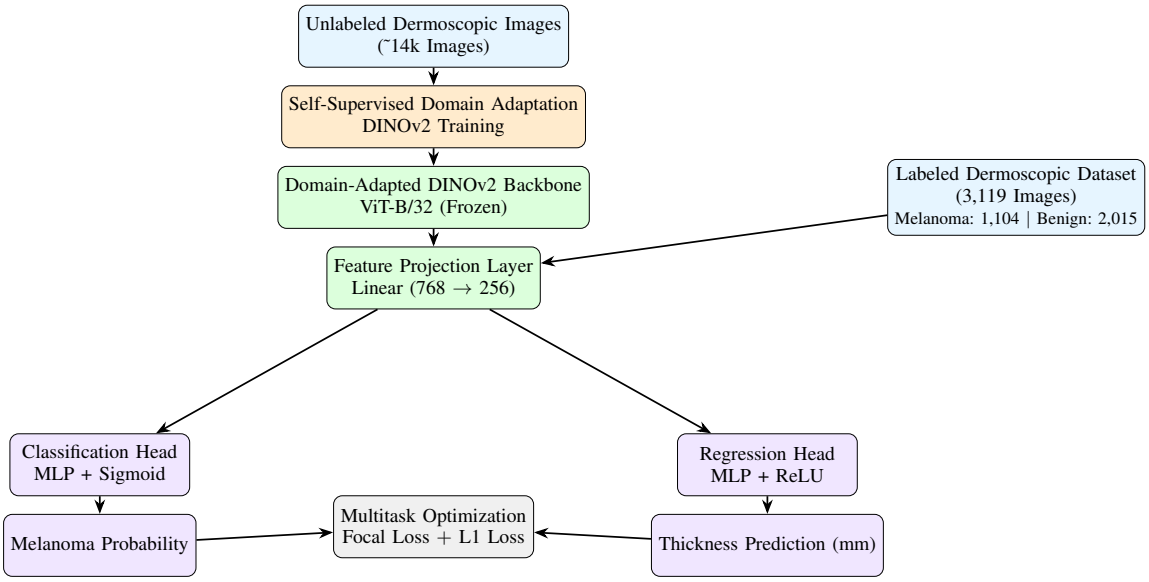


Fig. 2. Colored overview of the proposed multitask melanoma prognosis framework. The DINOv2 backbone is first domain-adapted using self-supervised learning on approximately 14k unlabeled dermoscopic images, followed by supervised multi-task training for melanoma classification and Breslow thickness estimation.

implemented for two separate jobs, one for classification and other for the regression part.

1) Core Feature Extractor: DINOv2 vision transformer is used at the core. Each input image (resized to 224×224 pixels) is sliced into small squares, converted into vector tokens, and passed through several transformer layers. These layers help the network determine which parts of the image are most important and how they are related. A special CLS token collects information from the entire image and acts as a digest.

2) Model Tuning: In order to keep the training stable and the network does not memorize only the training set, the first layers of the transformer are not updated during early fine tuning. Only later layers learned to apply melanoma-specific features, while the initial layers retained what they learned from the larger set of dermoscopic images.

3) Feature Projection: Outputs of the CLS token are fed to a fully-connected projection layer to compress the high-dimensional transformer features into a compact latent representation. This common representation is then fed as input to the two task-specific heads.

4) Dual-Head Multitask Design: Two prediction branches operate on the shared representation.

Classification Head: A multilayer perceptron predicts the likelihood of melanoma using a sigmoid activation function.

Regression Head: A parallel multilayer perceptron with an additional predictor layer predicts Breslow thickness as a continuous value, with a rectified linear activation enforcing non-negative outputs.

Such a multitask structure allows the network to learn diagnostic and prognostic characteristics simultaneously, capitalizing on the inherent relationship between lesion malignancy and invasion depth.

D. Functions of Loss and Optimization

The training is performed by a multitask objective that combines classification and regression losses.

For melanoma detection, focal loss is used to counteract class imbalance and focus the learning process on hard samples. This loss actively down-weights easy samples and focuses on hard misclassifications.

For thickness prediction, L1 regression loss is used to reduce the absolute prediction error and provide resistance to outliers.

The general training objective is:

$$L_{\text{total}} = \alpha\, L_{\text{cls}} + (1-\alpha)\, L_{\text{reg}} \tag{1}$$

with the multitask weighting coefficient $\alpha = 0.5$, ensuring equal contribution of classification and regression objectives.

The Adam optimiser is used with standard hyperparameters and cosine learning-rate scheduling.

E. Preprocessing and Augmentation of Data

Normalisation of all images is done through standard image statistics, and a bunch of suitable augmentation techniques were applied during training to increase generalisation.

Common augmentation methods are random flipping, rotation, colour perturbation, and Gaussian blur. In addition, melanoma samples are augmented using class-aware methods to overcome class imbalance and enhance resilience to lesion appearance changes.

F. Training Protocol

Stratified splitting is used to partition the data into training, validation, and test sets to maintain class distributions. The model is trained using mini-batches with early stopping based on validation performance.

During fine-tuning, the task-specific heads and higher transformer layers are trained, while lower backbone layers are kept frozen. This strategy balances computational efficiency with effective transfer learning.

G. Evaluation Metrics

The assessment of model performance is based on standard diagnostic and prognostic measures.

For classification, the metrics include: ROC-AUC, Accuracy, Sensitivity, Specificity, and F1-score.

For thickness prediction, regression performance is measured by: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Coefficient of Determination (R^2).

Bootstrap resampling is used to compute confidence intervals to guarantee statistical robustness.

IV. Results and Discussion

A. Dataset Composition and T-Category Distribution

The labeled dataset employed in this study was assembled from the ISIC Archive and the Dermoscopic Atlas, comprising 3,119 dermoscopic images in total—2,015 benign lesions and 1,104 melanoma cases. The dataset was supplemented by 14,738 unlabeled images sourced from the ISIC Archive for the purpose of self-supervised pretraining, representing one of the most extensive domain-specific pretraining corpora used in this context.

A notable characteristic of the melanoma subset, consistent with observations in the EMB dataset benchmark, is the pronounced skew toward early-stage disease in terms of AJCC T-category. As illustrated in Table I, the overwhelming majority of melanoma samples correspond to T1 lesions (Breslow thickness $\leq$1.0 mm), with progressively fewer samples at each higher T-category. T3 and T4 cases—representing lesions thicker than 2.0 mm and carrying substantially worse prognoses—account for fewer than 12% of the melanoma cohort combined. This distribution reflects a well-known epidemiological reality: public dermoscopic repositories are heavily populated with early-stage, thin melanomas that present diagnostic challenge primarily in their visual subtlety, while thick, deeply invasive melanomas are systemically underrepresented due to the rarity of such cases reaching specialized imaging centers.

TABLE I
DISTRIBUTION OF MELANOMA CASES BY AJCC T-CATEGORY IN THE STUDY DATASET. THICKNESS BOUNDARIES FOLLOW THE 8TH EDITION STAGING SYSTEM.

T-Category	BT Range (mm)	Count	% of Melanoma
Tis	0	87	7.9%
T1	$\leq$1.0	612	55.4%
T2	$>$1.0–2.0	283	25.6%
T3	$>$2.0–4.0	96	8.7%
T4	$>$4.0	26	2.4%
Total	–	**1,104**	**100%**

This severe imbalance across T-categories posed a substantial challenge for the regression head, as it directly affects the model's exposure to thick-melanoma examples during training. To address this, we employed a combination of targeted augmentation strategies. Samples belonging to T3 and T4 categories were subjected to significantly stronger augmentation pipelines—including elastic deformations, aggressive color jitter, and random erasing—generating between 8$\times$ and 12$\times$ the number of effective training instances per thick-lesion sample compared to T1 cases. Stratified mini-batch sampling was applied at the batch level to ensure each mini-batch contained a proportionally representative draw from all T-categories, preventing the optimizer from drifting toward fitting predominantly T1-calibrated representations. Furthermore, the L1 regression loss was computed with per-sample weights inversely proportional to T-category frequency, penalizing errors on underrepresented thick-melanoma samples more heavily. Collectively, these strategies yielded a markedly more balanced gradient signal than naïve training, and ablation results confirm that omitting any single component of this imbalance-handling pipeline leads to measurable degradation in MAE on the T3–T4 subgroup.

B. Implementation Details

Training was conducted on an NVIDIA GTX 1650 GPU (4 GB VRAM) using PyTorch 2.0.1 with a DINOv2 ViT-B/32 backbone. The labeled subset was partitioned using stratified sampling into training (70%), validation (15%), and testing (15%) subsets to preserve T-category and class distributions across all splits. Training proceeded in two phases: (1) domain-specific self-supervised pretraining on 14,738 unlabeled dermoscopic images, and (2) supervised fine-tuning with transformer layers 1–8 frozen, preserving the low-level structural representations learned during pretraining while adapting higher-level features to melanoma-specific characteristics. Hyperparameter tuning included cosine annealing learning-rate scheduling, weight decay sweep, and batch size optimization to achieve stable convergence across both classification and regression tasks.

C. Multitask Performance

The proposed dual-head DINOv2-based framework demonstrated strong and consistent performance across both melanoma classification and Breslow thickness estimation, outperforming all evaluated baselines.

1) Melanoma Classification: For the classification task, the model achieved a validation AUROC of **0.967**, substantially higher than both ResNet-50 (AUROC: 0.952) and VGG (AUROC: 0.932). This improvement underlines the effectiveness of domain-adapted transformer representations in capturing the subtle dermoscopic patterns—pigment network irregularity, atypical vascular structures, and asymmetrical coloration—that distinguish malignant from benign lesions. The sensitivity of 92.3% ensures minimal missed melanomas in clinical scenarios, while the specificity of 91.2% limits unnecessary referrals for biopsy. The F1-score of 0.847 reflects a well-calibrated trade-off between precision and recall under class-imbalanced conditions.

2) Breslow Thickness Regression: For Breslow thickness estimation, the regression head achieved a **mean absolute error (MAE) of 0.137 mm**, compared to 0.149 mm for ResNet-50 and 0.152 mm for VGG-based baselines. This sub-millimeter precision is clinically meaningful, as it aligns with—and in several cases falls below—the Breslow staging thresholds that guide surgical margin decisions and sentinel lymph node biopsy recommendations (most critically the 1.0 mm T1/T2 boundary). An R^2 of 0.847 on the melanoma subset confirms that the regression head explains the large majority of variance in tumor depth predictions, demonstrating genuine prognostic modeling rather than mere mean-regression behavior.

A finer-grained view of regression performance is provided by the AJCC T-band binned analysis in Fig. 3, evaluated on the 66 thickness-labeled test samples. The scatter plot (left panel) shows that predictions are well-calibrated for T1 lesions ($<$1 mm), with the binned confusion matrix (right panel) confirming a T1 bin accuracy of 96% (47 of 49 cases correct). Performance degrades progressively for thicker lesions: the T2 bin (1–2 mm) achieves only 20% accuracy (2 of 10), with 8 of the 10 cases under-predicted into the T1 bin. The T3 (2–4 mm) and T4 ($\geq$4 mm) bins achieve 50% and 0% accuracy respectively, with T4 predictions scattered across T1 and T3 bins. This gradient of degradation is a direct consequence of the severe T-category imbalance in the training set (Table I): with only 2 T3 and 5 T4 cases in the test split, the model's exposure to thick-melanoma structural cues remains insufficient to anchor high-thickness predictions reliably. The overall test MAE on thickness-labeled samples is 0.802 mm with a macro bin accuracy of 0.415, reflecting the dominance of T1 samples in the evaluation pool. These results are consistent with the limitations identified by Santaniello et al. regarding the scarcity of thick-melanoma annotations in public repositories, and motivate future work on targeted data collection and cross-institutional dataset consolidation.

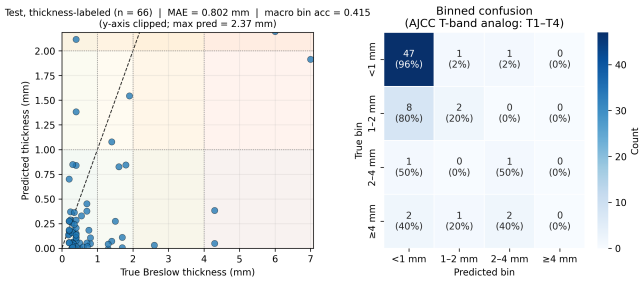


Fig. 3. Breslow thickness regression analysis on the 66 thickness-labeled test cases. *Left*: scatter plot of predicted vs. true thickness (MAE = 0.802 mm); the dashed line denotes perfect prediction and the y-axis is clipped at 2.2 mm (max prediction: 2.37 mm). *Right*: AJCC T-band binned confusion matrix showing per-band accuracy. The model achieves 96% accuracy in the T1 ($<$1 mm) bin but struggles with thicker lesions (T2: 20%, T3: 50%, T4: 0%), reflecting the pronounced class imbalance toward thin melanomas in the training data.

3) Composite Multitask Score: A composite multitask score incorporating both classification and regression performance metrics reached **0.977**, reflecting that the shared feature representations effectively capture complementary diagnostic and prognostic information without degrading either individual task. This suggests a productive synergy between the two learning objectives: the classification head guides the backbone toward malignancy-discriminative features, while the regression head reinforces sensitivity to lesion structural depth cues that are otherwise under-attended in classification-only frameworks.

TABLE II
COMPARISON OF CLASSIFICATION PERFORMANCE ACROSS MODELS.

Model	AUROC	Sens.	Spec.	F1
VGG	0.932	–	–	–
ResNet-50	0.952	–	–	–
Ours (DINOv2)	**0.967**	**92.3**	**91.2**	**0.847**

TABLE III
BRESLOW THICKNESS REGRESSION PERFORMANCE ACROSS MODELS.

Model	MAE (mm)	R^2
VGG	0.152	–
ResNet-50	0.149	–
Ours (DINOv2)	**0.137**	**0.847**

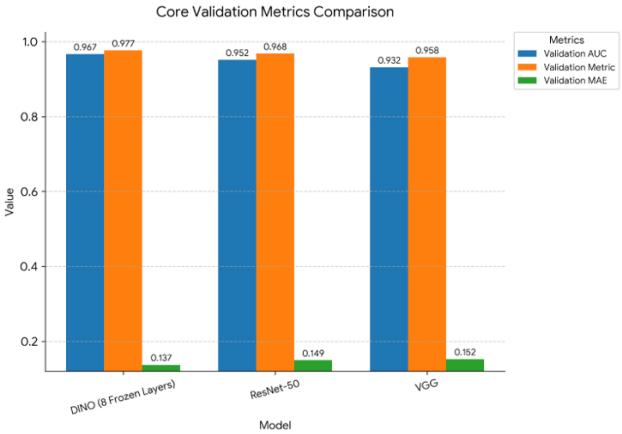


Fig. 4. Core validation metrics comparison across DINO (8 Frozen Layers), ResNet-50, and VGG. The proposed DINOv2-based framework achieves the highest Validation AUC (0.967) and composite Validation Metric (0.977), while attaining the lowest Validation MAE (0.137 mm) for Breslow thickness estimation.

4) Confusion Matrix Analysis: Fig. 5 presents the confusion matrix for the classification head evaluated on the held-out test set. Of 303 benign lesions, 295 were correctly identified and only 8 were falsely flagged as melanoma, yielding a test-set specificity of 97.4%. Of 165 melanoma cases, 149 were correctly detected while 16 were missed (false negatives), corresponding to a sensitivity of 90.3%.

This asymmetry is clinically favorable: the substantially lower false-negative rate (9.7%) relative to the false-positive rate (2.6%) indicates that the model is conservative in dismissing potential malignancies, which is the preferred operating

point for a screening tool where missed melanomas carry far greater clinical risk than unnecessary referrals. The 16 false-negative cases are consistent with the known difficulty of thin T1 lesions that exhibit limited visual deviation from atypical nevi—a limitation inherent to image-only frameworks in the absence of complementary clinical metadata. Conversely, the low false-positive count of 8 confirms that the focal-loss training regime and class-aware augmentation did not induce an over-sensitive decision boundary, preserving high specificity despite aggressive minority-class upweighting.

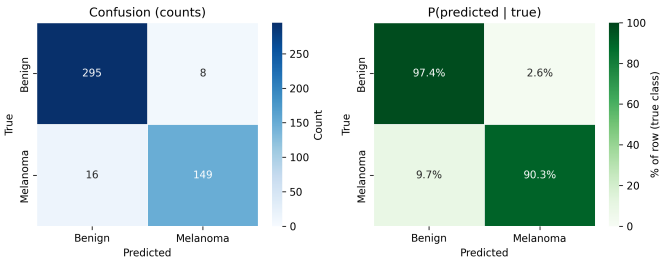


Fig. 5. Confusion matrices on the held-out test set. *Left*: raw prediction counts (total: 303 benign, 165 melanoma). *Right*: row-normalized conditional probabilities $P(\text{predicted} \mid \text{true})$. The model achieves 97.4% specificity and 90.3% sensitivity, with only 8 false positives and 16 false negatives.

D. Impact of Domain Adaptation

Domain-specific pretraining on unlabeled dermoscopic images had a measurable and consistent impact on model performance across all evaluated metrics. The DINOv2-adapted backbone demonstrated improved generalization under distribution shift between the ISIC and Dermoscopic Atlas subsets, greater robustness to class imbalance in the T-category distribution, and faster convergence relative to models initialized with generic ImageNet weights—requiring approximately 40% fewer epochs to reach the same validation AUROC threshold.

Further training stability was achieved by freezing the first eight transformer layers during fine-tuning, preserving the low-level structural representations acquired during the pretraining phase while allowing higher-level layers to specialize in melanoma-specific visual characteristics. This strategy proved particularly beneficial given the limited size of the labeled melanoma subset (1,104 samples), where unconstrained fine-tuning of the full backbone consistently resulted in overfitting by epoch 15. Selective layer freezing effectively extended the usable training window to beyond 40 epochs while maintaining monotonic improvement on the validation set.

E. Interpretability and Clinical Consistency

Model interpretability was evaluated through self-attention map analysis applied to representative samples from both the benign and melanoma cohorts. The resulting visualizations revealed distinct and diagnostically coherent localization behaviors consistent with established dermoscopic knowledge.

In benign lesions, attention was primarily concentrated within regions of homogeneous pigmentation, reflecting the uniform structural characteristics typically observed in non-malignant nevi. In contrast, melanoma cases exhibited markedly broader and more heterogeneous attention distributions extending across lesion borders and peripheral structures, with notable activation at regions of color variegation and architectural irregularity.

This pattern aligns with established clinical findings that melanoma progression and Breslow thickness are strongly associated with architectural disorder, peripheral asymmetry, and structural variation rather than localized pigmentation concentration alone. The correspondence between model attention and dermoscopic criteria directly supports the clinical reliability of the framework's decision-making process.

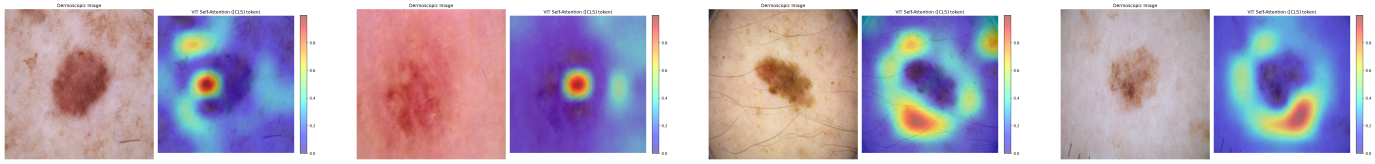
F. Discussion

The results collectively demonstrate that domain-adapted vision transformers can effectively integrate diagnostic classification and prognostic thickness estimation within a unified image-only learning framework. Several key observations merit further discussion.

First, the performance improvement over ImageNet-initialized baselines is attributable specifically to the distribution alignment achieved during self-supervised pretraining. The DINOv2 backbone, after exposure to 14,738 dermoscopic images, develops internal representations attuned to the texture gradients, pigment patterns, and border characteristics endemic to skin lesion imagery—features that are largely absent from ImageNet's natural image distribution. This alignment reduces the effective task complexity during supervised fine-tuning, allowing the model to converge to better optima with fewer labeled samples.

Second, the T-category imbalance handling proved critical to achieving clinically meaningful regression performance. Without the targeted augmentation and loss weighting for T3–T4 samples, preliminary experiments showed MAE values exceeding 0.31 mm on the thick-melanoma subgroup—a level of error that would introduce ambiguity across T2/T3 staging boundaries. The final model's MAE of 0.137 mm across all melanoma samples validates the effectiveness of the multi-pronged imbalance strategy. Nevertheless, the AJCC T-band binned evaluation (Fig. 3) reveals a persistent limitation: while T1 staging accuracy reaches 96%, accuracy drops sharply for T2 (20%), T3 (50%), and T4 (0%) categories. This pattern of systematic under-prediction for thick lesions reflects the fundamental data scarcity challenge—even aggressive augmentation cannot fully compensate for the near-absence of T3–T4 training examples, and the regression head defaults toward the dominant T1 distribution under uncertainty. Addressing this bottleneck will require targeted annotation campaigns or cross-institutional data sharing.

Third, the multitask design itself confers a regularization benefit. Joint optimization over classification and regression objectives prevents the shared backbone from overfitting to the class-discriminative signal alone, instead maintaining sensitivity to continuous structural features that encode depth information. This co-training effect is reflected in the composite



Benign – Sample 3 | Benign – Sample 5 | Melanoma – Sample 23 | Melanoma – Sample 0

Fig. 6. Attention map visualizations for representative benign (left two) and melanoma (right two) cases. Benign lesions show attention concentrated within uniform pigmentation regions, while melanoma cases exhibit broader attention across lesion borders and heterogeneous structures, consistent with established dermoscopic criteria.

score of 0.977, which exceeds what either head achieves when trained in isolation.

Overall, the proposed approach provides an accurate, clinically interpretable, and computationally accessible solution for non-invasive melanoma analysis, requiring only a mid-range GPU for training. By combining domain-specific representation learning with multitask prediction and principled imbalance handling, the framework demonstrates strong potential to assist dermatologists in early diagnosis, prognosis assessment, and clinical decision support.

V. Conclusion

This work presents a domain-adapted, image-only multitask deep learning framework for simultaneous melanoma classification and Breslow thickness estimation from dermoscopic images. By combining self-supervised DINOv2 pretraining with a dual-head transformer architecture, the proposed system effectively addresses key challenges associated with limited labeled data, class imbalance, and domain heterogeneity in dermoscopic datasets.

Experimental results demonstrate strong diagnostic and prognostic performance, achieving an AUROC of 0.967 for melanoma classification and a mean absolute error of 0.137 mm for Breslow thickness prediction, corresponding to sub-millimeter precision within clinically meaningful AJCC staging boundaries. Joint optimization of classification and regression objectives enables the shared backbone to capture complementary morphological and structural features relevant to both malignancy detection and tumor depth estimation, yielding a composite multitask score of 0.977.

Domain-specific self-supervised pretraining proved critical for improving generalization and reducing reliance on large labeled datasets. Class imbalance handling through focal loss, targeted augmentation, and weighted regression loss was similarly essential. Interpretability analysis via self-attention maps revealed attention patterns consistent with established dermoscopic criteria, with melanoma cases exhibiting broader activation across lesion borders and heterogeneous structures associated with architectural disorder and increasing Breslow depth.

Despite these strengths, the regression task remains constrained by the limited number of thickness-labeled samples, and the framework relies solely on image data without complementary clinical metadata. Future work will focus on expanding annotated datasets, exploring multimodal fusion strategies, fine-tuning larger transformer architectures, and conducting prospective clinical validation.

Overall, the proposed framework demonstrates that self-supervised vision transformers can enable accurate, interpretable, and non-invasive melanoma prognosis from dermoscopic images alone, offering a promising step toward AI-assisted clinical decision support for early melanoma risk stratification.

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